

BIG RAPIDS ROBEN-HOOD, MI, USA

WMO: 725416

Lat: 43.717N

Lon: 85.500W

Elev: 928

StdP: 14.21

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 14864

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -5.7	(c) 0.9	(d) -10.1	(e) 3.3	(f) -4.4	(g) -4.4	(h) 4.5	(i) 3.8	(j) 20.9	(k) 24.3	(l) 19.1	(m) 23.8	(n) 1.4	(o) 270	(p) 0.535

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.5	88.8	70.9	85.7	69.8	82.3	68.3	74.7	83.0	72.9	81.2	71.3	79.0	8.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.3	123.6	79.0	70.4	115.8	77.0	68.9	109.9	75.8	38.7	83.3	37.1	81.1	35.7	78.8	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.5	17.2	15.4	-14.9	92.3	7.1	3.3	-20.0	94.6	-24.2	96.5	-28.2	98.4	-33.4	100.7
			-15.3	77.6	7.1	2.5	-20.4	79.4	-24.5	80.8	-28.5	82.2	-33.6	84.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.0	21.7	22.4	32.0	44.4	55.4	65.0	69.1	67.2	60.0	48.4	37.6	27.3
	DBStd	18.97	10.98	10.68	11.31	9.08	8.37	6.79	6.13	5.68	7.52	8.52	8.68	9.83
	HDD50	3703	878	772	569	216	37	0	0	0	7	138	382	704
	HDD65	7346	1343	1192	1024	621	317	84	29	42	185	519	822	1168
	CDD50	2244	0	0	11	47	205	449	591	533	309	88	10	1
	CDD65	410	0	0	1	2	20	83	154	110	36	4	0	0
	CDH74	4535	0	0	9	45	276	955	1698	1102	413	37	0	0
	CDH80	1285	0	0	1	8	60	291	544	282	95	4	0	0
(14) Wind	WSAvg	5.9	6.6	6.7	6.4	7.2	6.3	5.3	4.7	4.4	4.6	5.7	6.2	6.4
(15) Precipitation	PrecAvg	33.2	2.0	1.5	2.3	3.0	3.1	3.1	2.6	3.8	3.7	2.8	3.0	2.2
	PrecMax	46.1	4.0	4.2	5.6	5.7	8.7	9.6	8.0	11.3	19.1	7.7	7.1	5.3
	PrecMin	20.4	0.6	0.3	0.3	0.6	0.6	0.8	0.6	0.4	0.0	0.5	0.4	0.5
	PrecStd	6.0	0.9	0.9	1.2	1.2	1.9	1.7	1.5	2.3	2.9	1.5	1.5	1.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.7	55.2	73.6	79.3	85.7	90.7	93.0	90.6	88.2	79.1	67.9	56.6
		MCWB	47.5	47.6	62.8	60.1	67.4	72.8	73.0	71.0	69.1	66.4	56.7	52.7
	2%	DB	45.0	46.2	61.5	73.0	81.3	87.6	89.7	86.3	82.3	73.0	61.2	48.1
		MCWB	40.7	42.0	52.3	56.3	65.3	70.1	71.7	70.3	68.2	61.6	52.9	45.3
	5%	DB	39.2	41.3	55.1	68.2	76.6	83.2	85.9	82.3	79.1	68.4	56.6	44.1
		MCWB	36.7	37.1	48.0	53.8	62.3	68.6	70.6	68.8	66.3	61.0	49.9	40.9
	10%	DB	36.2	36.7	49.0	62.9	72.3	79.6	82.2	80.7	75.0	63.9	52.3	39.4
		MCWB	34.5	33.8	42.4	50.3	59.9	65.8	68.7	67.4	64.6	57.2	47.2	37.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.4	50.3	62.4	64.3	71.0	75.3	78.2	76.0	74.0	70.2	59.1	53.9
		MCDB	49.4	52.6	72.4	75.2	79.8	85.4	87.8	84.0	80.3	76.8	64.1	56.1
	2%	WB	40.8	42.9	55.4	59.6	67.9	72.7	75.3	73.6	71.3	65.7	54.7	46.2
		MCDB	43.8	46.9	60.2	69.7	76.5	82.5	83.6	81.4	78.0	69.5	59.6	47.6
	5%	WB	37.1	37.2	49.2	55.8	65.6	70.8	73.3	72.0	69.3	61.9	51.1	41.4
		MCDB	39.3	40.3	54.5	64.0	72.7	79.3	81.7	79.4	75.1	66.8	55.9	43.9
	10%	WB	34.3	34.0	43.1	51.7	62.6	68.6	71.5	70.1	67.1	58.2	46.9	37.4
		MCDB	36.0	36.6	48.7	61.5	70.0	76.6	79.5	76.8	73.4	63.5	51.0	39.4
(35) Mean Daily Temperature Range	5% DB	MDBR	14.4	17.2	19.8	22.8	23.3	23.1	23.5	22.9	23.4	19.9	15.7	12.4
		MCDBR	16.1	20.2	28.3	33.0	28.5	27.7	26.5	25.8	28.0	27.3	24.2	17.2
	5% WB	MCWBR	13.8	15.7	19.5	18.0	14.4	13.1	11.9	12.7	15.2	17.3	17.0	14.8
		MCDWR	15.0	18.5	25.4	26.8	23.8	22.9	21.6	21.1	21.3	22.1	21.5	15.8
(40) Clear-Sky Solar Irradiance	taub		0.287	0.307	0.329	0.360	0.393	0.411	0.403	0.397	0.374	0.339	0.304	0.281
		taud	2.336	2.278	2.376	2.327	2.289	2.277	2.296	2.313	2.368	2.469	2.507	2.420
	Ebn at Noon		266	280	289	287	279	273	274	272	270	267	261	259
		Edh at Noon	25	32	34	38	41	42	40	38	34	27	22	22
(44) All-Sky Solar Radiation	RadAvg		400	679	1131	1447	1720	1916	1941	1646	1316	803	473	319
	RadStd		34	76	127	137	167	121	102	101	124	97	63	39

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page